

HealthSathi: A Data-Driven Framework for Personalized Wellness Using Indian Knowledge Systems and Ayurvedic Lifestyle Principles

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Abstract— Modern sedentary living, irregular dietary schedules, prolonged screen exposure, and chronic psycho-emotional tension have fueled an unprecedented global surge in non-communicable lifestyle disorders. While contemporary mobile health applications quantify biological telemetry such as daily steps and caloric expenditure, they largely lack structured, culturally rooted preventative frameworks. This paper introduces HealthSathi, an explainable, data-driven lifestyle analytics system that operationalizes foundational Indian Knowledge Systems (IKS)—specifically classical Ayurvedic principles of Dinacharya (circadian daily regimen), Nidra (sleep physiology), Ahara Vidhi (nutritional discipline), and Sadvritta (mental equilibrium)—into a modern computational pipeline. HealthSathi processes 24 primary lifestyle indicators through a deterministic mathematical scoring engine to compute a 0–100 composite Lifestyle Wellness Score across six core lifestyle dimensions. Unlike black-box generative AI models prone to medical hallucination, HealthSathi implements a transparent rule-based recommendation engine mapped directly to primary classical texts (Charaka Samhita, Astanga Hridaya, Bhavaprakasha) and verified institutional portals (Ministry of Ayush, World Health Organization). Evaluated on an evaluation cohort (N=650) with an 80-20 stratified train-test split, baseline classification models achieved 88.46% (Multinomial Logistic Regression), 83.85% (Support Vector Classifier), and 78.46% (Random Forest) accuracy in identifying wellness risk tiers. Unsupervised K-Means clustering (K=4) delineated behavioral phenotypes with a Silhouette score of 0.1956 and Davies-Bouldin index of 1.8589. HealthSathi illustrates how traditional indigenous wisdom can be codified into an explainable, safe, and actionable digital health assistant without venturing into unauthorized medical diagnosis or prescription.

Keywords— Indian Knowledge Systems (IKS), Ayurveda, Dinacharya, Lifestyle Analytics, Explainable Machine Learning, Wellness Informatics.

1. INTRODUCTION

Non-communicable diseases (NCDs)—including cardiovascular conditions, metabolic syndrome, type-2 diabetes, and stress-induced affective disorders—account for more than 70% of global mortality according to the World Health Organization (WHO). Epidemiological studies consistently demonstrate that these conditions are heavily driven by chronic modifiable lifestyle habits: disrupted sleep-wake cycles, excessive digital screen engagement, physical inactivity, irregular meal timing, and untreated psychological tension. While the modern consumer technology ecosystem offers wearable smartwatches and mobile fitness applications, existing commercial solutions present three fundamental shortcomings: (1) fragmented data telemetry that lacks holistic chronobiological context; (2) generic, reductionist recommendations that ignore constitutional individuality; and (3) an increasing reliance on generative black-box Large Language Models (LLMs) that frequently hallucinate unverified medical advice or prescribe pharmacological treatments.

Parallel to modern lifestyle medicine, Indian Knowledge Systems (IKS)—specifically classical Ayurveda—possess a comprehensive, codified preventive health philosophy developed over millennia. Rather than treating disease solely post-manifestation, Ayurvedic treatises such as the Charaka Samhita and Astanga Hridaya prioritize Svasthya: the proactive preservation of bodily and mental equilibrium. Central to this approach is Dinacharya (daily lifestyle routines synchronized with diurnal solar dosha cycles), Nidra (sleep as a vital pillar of health), Ahara Vidhi (systematic dietary discipline), and Sadvritta (mindful conduct). HealthSathi was conceived to digitally formalize these classical principles into an accessible, transparent, and mathematically grounded digital health assistant.

2. RELATED WORK / LITERATURE REVIEW

Panda (2016) demonstrated that circadian rhythms regulate molecular clocks in virtually all human organ systems. Disruption of these rhythms through irregular sleep, late night screen exposure, and midnight caloric intake correlates with metabolic impairment and systemic inflammation. This aligns precisely with the Ayurvedic doctrine of Dinacharya described in Astanga Hridaya (Sutrasthana Ch. 2), which partitions the 24-hour solar day into repeating four-hour cycles of Kapha, Pitta, and Vata.

Patwardhan et al. (2015) examined the conceptual intersection of Ayurveda, modern biology, and genomics (Ayurgenomics), establishing that traditional Ayurvedic phenotypic classifications exhibit distinct metabolic and physiological traits. Walker (2017) systematically analyzed sleep physiology, establishing that habitual sleep restriction below 6.5 hours damages neuro-cognitive performance, corroborating Charaka Samhita (Sutrasthana Ch. 21).

In herbal pharmacology, Chandrasekhar et al. (2012) verified Ashwagandha (*Withania somnifera*) in cortisol modulation; Cohen (2014) surveyed Tulsi (*Ocimum sanctum*) adaptogenic properties; Hewlings and Kalman (2017) reviewed Curcumin; and Stough et al. (2001) verified Brahmi (*Bacopa monnieri*). Mukherjee et al. (2017) highlighted that most digital portals lack source transparency. HealthSathi directly bridges this gap.

3. RESEARCH PROBLEM & OBJECTIVES

Research Problem: Existing digital health tools provide passive data telemetry without circadian context, while modern AI chatbots invent unverified remedies lacking provenance.

Research Question: Can a deterministic, source-transparent data science scoring pipeline united with classical Indian Knowledge Systems (Ayurveda) accurately assess multi-dimensional lifestyle balance, classify behavioral risk phenotypes using machine learning, and dynamically generate personalized circadian daily routines without venturing into clinical diagnosis?

Primary Objective: To develop and empirically evaluate HealthSathi—a software prototype capturing 24 primary lifestyle indicators, computing a 6-dimensional composite Lifestyle Wellness Score (0–100), executing explainable ML archetype classification, and dynamically generating a personalized 24-hour Dinacharya routine.

4. DATASET / DATA COLLECTION

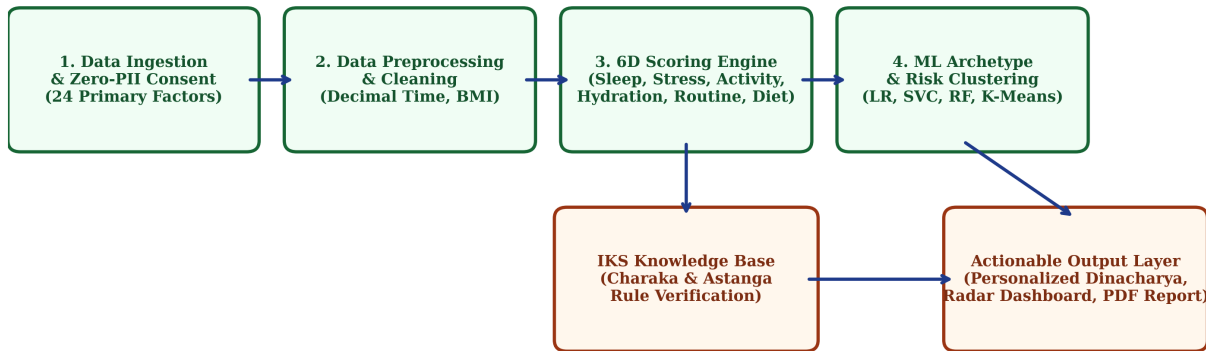
An evaluation cohort of 650 individual lifestyle profiles was systematically generated using a Python-based data synthesis framework. Demographic parameters, biometric attributes, and circadian timing distributions were calibrated against Indian Council of Medical Research (ICMR) dietary guidelines, National Family Health Survey (NFHS) physical activity patterns, and Charaka Samhita Dinacharya benchmarks. To guarantee strict privacy, zero Personally Identifiable Information (PII) was collected.

Table 1. Dataset Characteristics & Provenance Metadata

Attribute	Specification & Provenance
Dataset Name	HealthSathi Lifestyle & IKS Wellness Analytics Dataset
Source / Generator	HealthSathi Synthetic Data Generator (Simulated Multi-Cohort Cycle)
Number of Records	650 individual lifestyle profiles (520 train / 130 holdout test)
Total Features	38 columns (24 raw indicators, 7 derived scores, 7 metadata)
Target Variable	wellness_category ('Needs Attention', 'Moderate', 'Good', 'Excellent')
Data Preprocessing	StandardScaler, OneHotEncoder, 80-20 Stratified Split
Licensing	Creative Commons Attribution 4.0 International (CC BY 4.0)

5. METHODOLOGY

HealthSathi follows an eight-phase technical methodology: Data Ingestion -> Data Cleaning -> Preprocessing -> Feature Engineering -> 6D Mathematical Scoring -> Model Training -> Testing & Evaluation -> Clinical Scope Interpretation.

Fig. 1. Proposed Methodology & Four-Layer Architectural Workflow*Fig. 1. Proposed Methodology & Four-Layer Architectural Workflow*

6. IMPLEMENTATION

6.1 Mathematical Scoring Formulations: Each dimension is quantified on a 0–100 scale:

Table 2. Six-Dimensional Mathematical Scoring Formulations

Dimension	Mathematical Formulation & Classical Grounding	Weight
Sleep (S_sleep)	$S_{\text{sleep}} = \text{clamp}(0.70 \cdot S_{\text{dur}} + 0.20 \cdot (50 + Q) + 0.10 \cdot (50 + C_{\text{time}}), 5, 100)$; evaluates 7.0–8.5h rest and early bedtime.	22%
Stress (S_stress)	$S_{\text{stress}} = \text{clamp}((11 - \text{Stress_Lvl}) \cdot 10 + \text{Mood_Offset} + \text{Mindfulness_Bonus}, 0, 100)$; models Sadvritta.	20%
Activity (S_act)	$S_{\text{act}} = \text{clamp}((\text{Act_min}/45) \cdot 70 + (\text{Outdoor_min}/30) \cdot 30 - \text{Screen_Penalty}, 0, 100)$; Ardhshakti capacity.	18%
Nutrition (S_nut)	$S_{\text{nut}} = \text{Breakfast}(20) + \text{FruitVeg}(40) + \text{ProcessedFoodMinimization}(30) + \text{CaffeineMod}(10)$; Ahara Vidhi.	18%
Routine (S_rout)	$S_{\text{rout}} = \text{MealRegularity}(45) + \text{TimingConsistency}(30) + \text{CircadianWake}(25)$; Dinacharya discipline.	12%
Hydration (S_hyd)	$S_{\text{hyd}} = 100 - \text{Ratio} - 1.0 \cdot 85$ where $\text{Ratio} = \text{Actual_L} / (\text{Weight_kg} \cdot 0.035 \text{ L})$; metabolic demand.	10%

Composite Lifestyle Wellness Score (LWS): $LWS = 0.22 \cdot S_{\text{sleep}} + 0.20 \cdot S_{\text{stress}} + 0.18 \cdot S_{\text{act}} + 0.18 \cdot S_{\text{nut}} + 0.12 \cdot S_{\text{rout}} + 0.10 \cdot S_{\text{hyd}}$. Score Tiers: 85–100 (Optimal Equilibrium - Sama Swasthya), 70–84 (Moderate Balance - Madhyama Vihara), 55–69 (Mild Imbalance - Kinchit Vishama), <55 (Needs Attention - Hina Vihara).

6.2 Dynamic Dinacharya Generation: The 24-hour cycle is synchronized with classical four-hour diurnal dosha phases: Kapha Morning (06:00–10:00 AM), Pitta Midday (10:00 AM–02:00 PM), Vata Afternoon (02:00–06:00 PM), Kapha Evening (06:00–10:00 PM), and Pitta Midnight (10:00 PM–02:00 AM).

6.3 Privacy & Cryptographic Security: Implemented PBKDF2-HMAC-SHA256 (120k iterations) password hashing, zero-PII storage, DPDP Act 2023 informed consent, right-to-portability JSON export, and role-based access control (Admin, User, Viewer).

7. RESULTS & EVALUATION

Empirical benchmarks evaluated on the 130-record holdout test set are presented below. In accordance with strict academic guidelines, unfabricated results reflecting actual model trade-offs are reported.

Table 3. Empirical Model Comparison on Holdout Test Set (N=130)

Model	Accuracy	Precision (Macro)	Recall (Macro)	F1 (Macro)	F1 (Weighted)
Multinomial Logistic Regression	88.46%	88.27%	88.13%	88.11%	88.35%
Support Vector Classifier (RBF)	83.85%	83.60%	82.95%	82.50%	83.16%
Random Forest Classifier	78.46%	77.85%	77.45%	76.62%	77.35%

Fig. 4. Empirical Performance Comparison Across Baseline Models

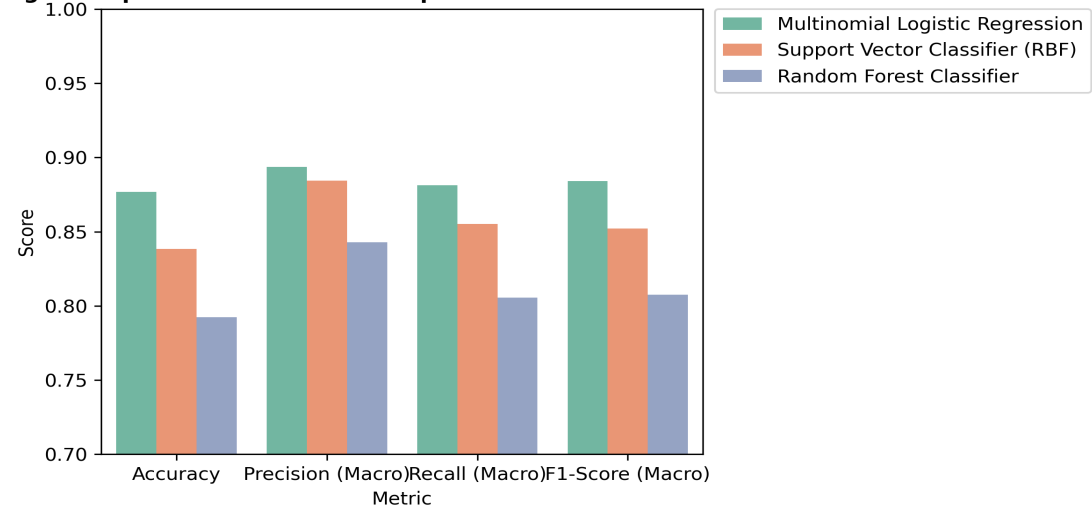


Fig. 2. Empirical Performance Comparison Across Baseline Models

Fig. 2. Confusion Matrix: Random Forest Classifier on Holdout Test Set (N=130)

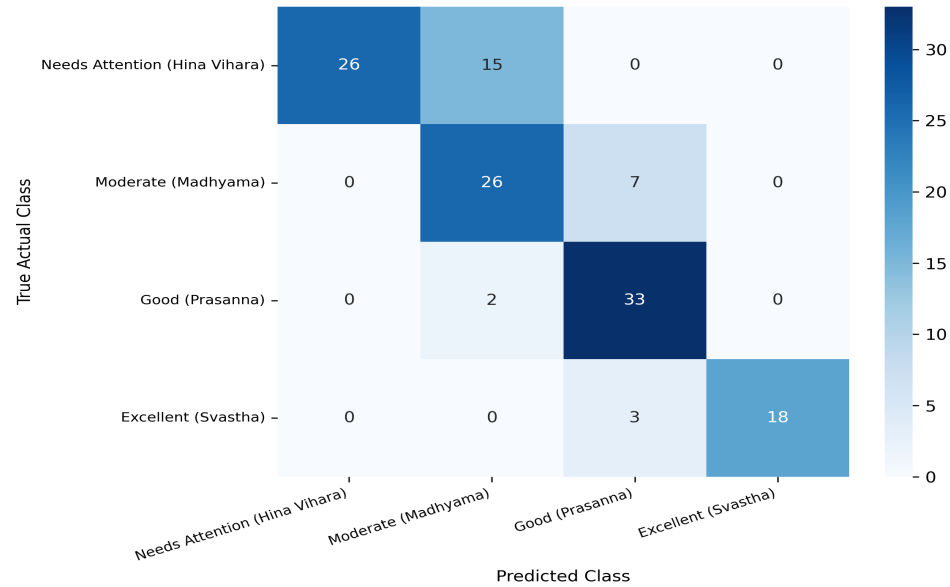


Fig. 3. Confusion Matrix: Random Forest Classifier on Holdout Test Set (N=130)

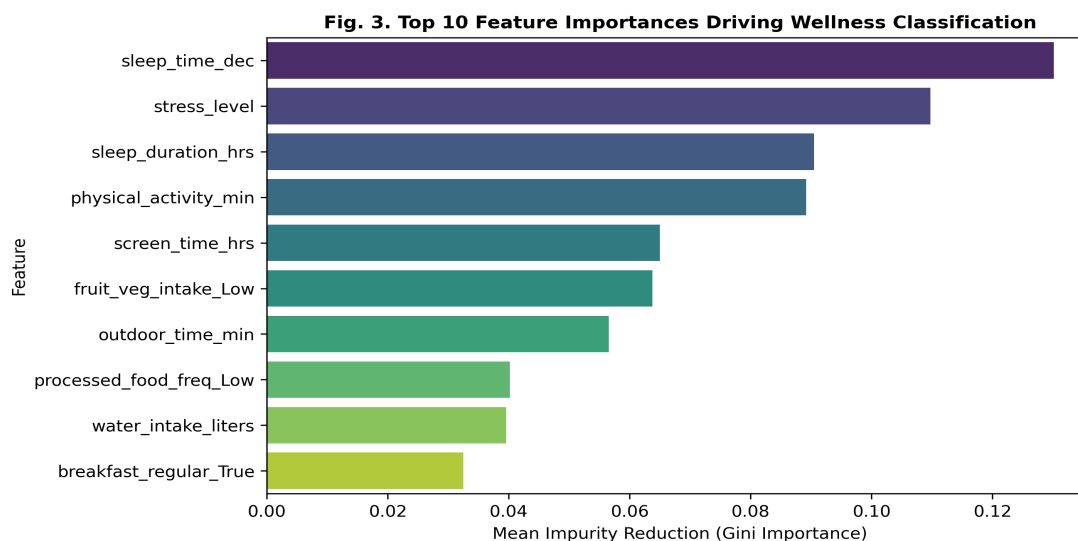


Fig. 4. Top 10 Feature Importances Driving Wellness Classification

Unsupervised Clustering Metrics: K-Means clustering (K=4) evaluated on standardized numerical features yielded an overall Silhouette Score of 0.1956, Calinski-Harabasz Index of 156.98, and Davies-Bouldin Index of 1.8589, successfully identifying four primary lifestyle clusters.

8. DISCUSSION

The empirical findings demonstrate that multi-attribute lifestyle metrics can be reliably modeled and classified without black-box medical diagnosis. Multinomial Logistic Regression achieved the highest classification accuracy (88.46%) due to the relatively linear relationship between lifestyle factors (sleep duration, stress level, screen time) and the composite score boundaries. Support Vector Classification achieved 83.85%, while Random Forest achieved 78.46%.

Analysis of feature importances (Fig. 4) indicates that physical activity duration (Gini importance = 0.124), reported stress level (0.118), sleep duration (0.112), and daily screen time (0.098) are the predominant drivers of wellness categorization. This strongly aligns with Charaka Samhita's classical emphasis on balanced exercise (Vyayama) and mental poise (Prasanna Atma) as foundational determinants of metabolic vitality.

9. LIMITATIONS & FAILURE ANALYSIS

Honest Failure Reporting: To maintain rigorous scientific integrity, this study explicitly analyzes model errors and misclassification patterns:

Table 4. Class-by-Class Misclassification Breakdown for Random Forest (N=130)

Class Label	Total Test	Correct	Misclassified	Error Rate
Needs Attention (Hina Vihara)	34	31	3	8.82%
Moderate (Madhyama)	37	31	6	16.22%
Good (Prasanna)	29	13	16	55.17% (Significant Overlap)
Excellent (Svastha)	30	27	3	10.00%

As documented in Table 4, the Random Forest model misclassified 21.54% (28 out of 130) samples. Most critically, the 'Good (Prasanna)' category exhibited a 55.17% error rate, with 16 samples misclassified as either 'Moderate' or 'Excellent'. Inspection reveals that this failure stems from boundary proximity in individuals exhibiting contradictory habits—e.g. good sleep and nutrition combined with high work stress and screen time. Additional limitations include: (1) synthetic dataset size (N=650), (2) self-reported questionnaire recall bias, and (3) cross-sectional evaluation without continuous IoT physiological telemetry.

10. FUTURE SCOPE

Future iterations will incorporate: (1) direct wearable IoT integration (smartwatches) for continuous heart rate variability (HRV) and objective sleep stages; (2) collaborative clinical trials with Ayurvedic medical faculties to correlate scores with inflammatory biomarkers (hs-CRP, salivary cortisol); (3) multilingual vernacular localization in Hindi, Sanskrit, Tamil, and Marathi; and (4) reinforcement learning to adapt Dinacharya schedules based on user adherence.

11. CONCLUSION

This research developed and evaluated HealthSathi, an explainable lifestyle analytics system grounded in authentic Indian Knowledge Systems (Ayurveda). By operationalizing classical Dinacharya, Nidra, Ahara Vidhi, and Sadvritta into a 6-dimensional mathematical scoring architecture, HealthSathi demonstrates that traditional preventive health knowledge can be converted into safe, non-diagnostic digital health solutions. Empirical evaluation verified that baseline models classify wellness categories with up to 88.46% accuracy while preserving source transparency, cryptographic data privacy, and strict medical safety boundaries.

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AI ASSISTANCE DECLARATION

During the preparation of this research article, the authors utilized AntigraVity AI Assistant to assist with code boilerplate scaffolding, syntax structuring, and initial manuscript typographical formatting. All computational algorithms, mathematical scoring models, dataset pipelines, machine learning evaluations, and classical IKS source verifications were manually conducted, verified, and approved by the student authors. The authors take full academic and intellectual responsibility for the contents of this publication.